



$$I(J^P) = 0(0^-)$$

I, J, P need confirmation.

Quantum numbers shown are quark-model predictions.

B_c^+ MASS

VALUE (MeV)	DOCUMENT ID	TECN	COMMENT
6274.47 ± 0.27 ± 0.17	¹ AAIJ	20R LHCb	pp at 7, 8, 13 TeV
• • • We do not use the following data for averages, fits, limits, etc. • • •			
6274.28 ± 1.40 ± 0.32	² AAIJ	17L LHCb	Repl. by AAIJ 20R
6274.0 ± 1.8 ± 0.4	³ AAIJ	14AQ LHCb	Repl. by AAIJ 20R
6276.28 ± 1.44 ± 0.36	⁴ AAIJ	13AS LHCb	Repl. by AAIJ 20R
6273.7 ± 1.3 ± 1.6	⁵ AAIJ	12AV LHCb	Repl. by AAIJ 20R
6275.6 ± 2.9 ± 2.5	⁶ AALTONEN	08M CDF	$p\bar{p}$ at 1.96 TeV
6300 ± 14 ± 5	⁶ ABAZOV	08T D0	$p\bar{p}$ at 1.96 TeV
6285.7 ± 5.3 ± 1.2	⁶ ABULENCIA	06C CDF	Repl. by AALTONEN 08M
6400 ± 390 ± 130	⁷ ABE	98M CDF	$p\bar{p}$ at 1.8 TeV
6320 ± 60	⁸ ACKERSTAFF	980 OPAL	$e^+e^- \rightarrow Z$

¹ AAIJ 20R uses the $B_c^+ \rightarrow J/\psi \pi^+, J/\psi \pi^+ \pi^- \pi^+, J/\psi p\bar{p} \pi^+, J/\psi D_s^+, J/\psi D^0 K^+$ and $B_s^0 \pi^+$ modes.

² Measured using $B_c^+ \rightarrow J/\psi D^0 K^+$ decays.

³ Uses $B_c^+ \rightarrow J/\psi p\bar{p} \pi^+$ decays.

⁴ AAIJ 13AS uses the $B_c^+ \rightarrow J/\psi D_s^+$.

⁵ AAIJ 12AV uses the $B_c^+ \rightarrow J/\psi \pi^+$ mode and also measures the mass difference $M(B_c^+) - M(B^+) = 994.6 \pm 1.3 \pm 0.6 \text{ MeV}/c^2$.

⁶ Measured using a fully reconstructed decay mode of $B_c \rightarrow J/\psi \pi$.

⁷ ABE 98M observed $20.4^{+6.2}_{-5.5}$ events in the $B_c^+ \rightarrow J/\psi(1S) \ell \nu_\ell$ with a significance of > 4.8 standard deviations. The mass value is estimated from $m(J/\psi(1S) \ell)$.

⁸ ACKERSTAFF 980 observed 2 candidate events in the $B_c^+ \rightarrow J/\psi(1S) \pi^+$ channel with an estimated background of 0.63 ± 0.20 events.

$m_{B_c^+} - m_{B_s^0}$

VALUE (MeV)	DOCUMENT ID	TECN	COMMENT
907.75 ± 0.37 ± 0.27	¹ AAIJ	20R LHCb	pp at 7, 8, 13 TeV

¹ AAIJ 20R uses the $B_c^+ \rightarrow J/\psi \pi^+, J/\psi \pi^+ \pi^- \pi^+, J/\psi p\bar{p} \pi^+, J/\psi D_s^+, J/\psi D^0 K^+$ and $B_s^0 \pi^+$ modes.

B_c^+ MEAN LIFE

VALUE (10^{-12} s)	DOCUMENT ID	TECN	COMMENT
0.510 ± 0.009 OUR EVALUATION	(Produced by HFLAV)		
0.510 ± 0.009 OUR AVERAGE			
0.541 ± 0.026 ± 0.014	¹ SIRUNYAN	18BY CMS	pp at 8 TeV
0.5134 ± 0.0110 ± 0.0057	^{2,3} AAIJ	15G LHCb	pp at 7, 8 TeV
0.509 ± 0.008 ± 0.012	⁴ AAIJ	14G LHCb	pp at 8 TeV

0.452 ± 0.048 ± 0.027	³ AALTONEN	13	CDF	$p\bar{p}$ at 1.96 TeV
0.448 ± 0.038 ± 0.032 -0.036	⁵ ABAZOV	09H	D0	$p\bar{p}$ at 1.96 TeV
0.463 ± 0.073 ± 0.036 -0.065	⁵ ABULENCIA	06O	CDF	$p\bar{p}$ at 1.96 TeV
0.46 ± 0.18 ± 0.03 -0.16	⁵ ABE	98M	CDF	$p\bar{p}$ 1.8 TeV

¹ The lifetime is measured using the decays $B_c^+ \rightarrow J/\psi \pi^+$ and $B^+ \rightarrow J/\psi K^+$.

² Also measures the width difference $\Delta\Gamma = \Gamma_{B_c^+} - \Gamma_{B^+} = 4.46 \pm 0.14 \pm 0.07 \text{ mm}^{-1} \text{ c}$.

³ Uses fully reconstructed $B_c^+ \rightarrow J/\psi \pi^+$ decays.

⁴ Measured using $B_c^+ \rightarrow J/\psi \mu^+ \nu_\mu X$ decays.

⁵ The lifetime is measured from the $J/\psi e$ decay vertices.

B_c^+ DECAY MODES $\times B(\bar{b} \rightarrow B_c)$

The following quantities are not pure branching ratios; rather the fractions $\Gamma_i/\Gamma \times B(\bar{b} \rightarrow B_c)$. B_c^- modes are charge conjugates of the modes below.

Mode	Fraction (Γ_i/Γ)	Confidence level
Γ_1 $J/\psi(1S)\ell^+\nu_\ell$ anything	seen	
Γ_2 $J/\psi(1S)\mu^+\nu_\mu$	seen	
Γ_3 $J/\psi(1S)\tau^+\nu_\tau$	seen	
Γ_4 $J/\psi(1S)\pi^+$	seen	
Γ_5 $J/\psi(1S)K^+$	seen	
Γ_6 $J/\psi(1S)\pi^+\pi^0$		
Γ_7 $J/\psi(1S)\pi^+\pi^+\pi^-$	seen	
Γ_8 $J/\psi(1S)K^+\pi^-\pi^+$		
Γ_9 $J/\psi(1S)K^+K^-K^+$		
Γ_{10} $J/\psi(1S)a_1(1260)$	not seen	
Γ_{11} $J/\psi(1S)K^+K^-\pi^+$	seen	
Γ_{12} $J/\psi(1S)\pi^+\pi^+\pi^+\pi^-\pi^-$	seen	
Γ_{13} $\psi(2S)\pi^+$	seen	
Γ_{14} $\psi(2S)\pi^+$, $\psi(2S) \rightarrow J/\psi\pi^+\pi^-$		
Γ_{15} $\psi(2S)\pi^+\pi^-\pi^+$		
Γ_{16} $\psi(2S)K^+K^-\pi^+$		
Γ_{17} $J/\psi(1S)D^0K^+$	seen	
Γ_{18} $J/\psi(1S)D^*(2007)^0K^+$	seen	
Γ_{19} $J/\psi(1S)D^*(2010)^+K^{*0}$	seen	
Γ_{20} $J/\psi(1S)D^+K^{*0}$	seen	
Γ_{21} $J/\psi(1S)D_s^+$	seen	
Γ_{22} $J/\psi(1S)D_s^{*+}$	seen	
Γ_{23} $J/\psi(1S)p\bar{p}\pi^+$	seen	
Γ_{24} $\chi_{c0}\pi^+$		

Γ_{25}	$\chi_{c1} \pi^+$		
Γ_{26}	$\chi_{c2} \pi^+$		
Γ_{27}	$\chi_{c1}(3872) \pi^+, \chi_{c1}(3872) \rightarrow$ $J/\psi \pi^+ \pi^-$		
Γ_{28}	$p \bar{p} \pi^+$	not seen	
Γ_{29}	$D^0 K^+$	seen	
Γ_{30}	$D^0 \pi^+$	not seen	
Γ_{31}	$D^{*0} \pi^+$	not seen	
Γ_{32}	$D^{*0} K^+$	not seen	
Γ_{33}	$D_s^+ \bar{D}^0$	$< 7.2 \times 10^{-4}$	90%
Γ_{34}	$D_s^+ D^0$	$< 3.0 \times 10^{-4}$	90%
Γ_{35}	$D_s^+ \bar{D}^0$	$< 1.9 \times 10^{-4}$	90%
Γ_{36}	$D^+ D^0$	$< 1.4 \times 10^{-4}$	90%
Γ_{37}	$D_s^{*+} \bar{D}^0$	$< 5.3 \times 10^{-4}$	90%
Γ_{38}	$D_s^+ \bar{D}^*(2007)^0$	$< 4.6 \times 10^{-4}$	90%
Γ_{39}	$D_s^{*+} D^0$	$< 9 \times 10^{-4}$	90%
Γ_{40}	$D_s^+ D^*(2007)^0$	$< 6.6 \times 10^{-4}$	90%
Γ_{41}	$D^*(2010)^+ \bar{D}^0$	$< 3.8 \times 10^{-4}$	90%
Γ_{42}	$D^*(2010)^+ \bar{D}^0, D^{*+} \rightarrow$ $D^+ \pi^0 / \gamma$	not seen	
Γ_{43}	$D^+ \bar{D}^*(2007)^0$	$< 6.5 \times 10^{-4}$	90%
Γ_{44}	$D^*(2007)^+ D^0$	$< 2.0 \times 10^{-4}$	90%
Γ_{45}	$D^*(2010)^+ D^0, D^{*+} \rightarrow D^+ \pi^0 / \gamma$	not seen	
Γ_{46}	$D^+ D^*(2007)^0$	$< 3.7 \times 10^{-4}$	90%
Γ_{47}	$D_s^{*+} \bar{D}^*(2007)^0$	$< 1.3 \times 10^{-3}$	90%
Γ_{48}	$D_s^{*+} D^*(2007)^0$	$< 1.3 \times 10^{-3}$	90%
Γ_{49}	$D^*(2010)^+ \bar{D}^*(2007)^0$	$< 1.0 \times 10^{-3}$	90%
Γ_{50}	$D^*(2010)^+ D^*(2007)^0$	$< 7.7 \times 10^{-4}$	90%
Γ_{51}	$D^+ K^{*0}$	not seen	
Γ_{52}	$D^+ \bar{K}^{*0}$	not seen	
Γ_{53}	$D_s^+ K^{*0}$	not seen	
Γ_{54}	$D_s^+ \bar{K}^{*0}$	not seen	
Γ_{55}	$D_s^+ \phi$	not seen	
Γ_{56}	$K^+ K^0$	not seen	
Γ_{57}	$\pi^+ B^{*0}, B^{*0} \rightarrow \mu^+ \mu^-$		
Γ_{58}	$\pi^+ B_s^{*0}, B_s^{*0} \rightarrow \mu^+ \mu^-$		
Γ_{59}	$B_s^0 \pi^+ / B(\bar{b} \rightarrow B_s)$	seen	
Γ_{60}	$B_s^0 \pi^+$		
Γ_{61}	$\pi^+ \mu^+ \mu^-$		
Γ_{62}	$D_s^+ \mu^+ \mu^-$		

B_c^+ BRANCHING RATIOS **$\Gamma(J/\psi(1S)\ell^+\nu_\ell \text{ anything})/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)$** **$\Gamma_1/\Gamma \times B$**

VALUE (units 10^{-5})	CL%	DOCUMENT ID	TECN	COMMENT
8.2 ± 1.3 OUR AVERAGE		Error includes scale factor of 1.4.		
$8.8 \pm 1.0 \pm 0.2$		^{1,2} AALTONEN	16A CDF	$p\bar{p}$ at 1.96 TeV
$5.2^{+2.4}_{-2.1}$		³ ABE	98M CDF	$p\bar{p}$ 1.8 TeV

• • • We do not use the following data for averages, fits, limits, etc. • • •

<16	90	⁴ ACKERSTAFF	98O OPAL	$e^+e^- \rightarrow Z$
<19	90	⁵ ABREU	97E DLPH	$e^+e^- \rightarrow Z$
<12	90	⁶ BARATE	97H ALEP	$e^+e^- \rightarrow Z$

¹ AALTONEN 16A reports $[\Gamma(B_c^+ \rightarrow J/\psi(1S)\ell^+\nu_\ell \text{ anything})/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] / [B(B^+ \rightarrow J/\psi(1S)K^+)] = 0.211 \pm 0.012^{+0.021}_{-0.020}$ which we multiply by our best (shown rounded) values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = (40.8 \pm 0.7) \times 10^{-2}$, $B(B^+ \rightarrow J/\psi(1S)K^+) = (1.019 \pm 0.019) \times 10^{-3}$. Our first error is their experiment's error and our second error is the systematic error from using our best (shown rounded) values.

² AALTONEN 16A also measures the cross-section $\sigma(B_c) \times B(B_c \rightarrow J/\psi\mu\nu_\mu) = 0.60 \pm 0.09$ nb and estimates the total cross-section $\sigma(B_c)$ to be in the range 25 ± 4 to 52 ± 8 nb for $p_T(B_c) > 6$ GeV/c and $|y(B_c)| < 1$.

³ ABE 98M result is derived from the measurement of $[\sigma(B_c) \times B(B_c \rightarrow J/\psi(1S)\ell\nu_\ell)] / [\sigma(B^+) \times B(B^+ \rightarrow J/\psi(1S)K^+)] = 0.132^{+0.041}_{-0.037}(\text{stat}) \pm 0.031(\text{sys})^{+0.032}_{-0.020}(\text{lifetime})$ by using PDG 98 values of $B(b \rightarrow B^+)$ and $B(B^+ \rightarrow J/\psi(1S)K^+)$.

⁴ ACKERSTAFF 98O reports $B(Z \rightarrow B_c X)/B(Z \rightarrow qq) \times B(B_c \rightarrow J/\psi(1S)\ell\nu_\ell) < 6.95 \times 10^{-5}$ at 90%CL. We rescale to our PDG 98 values of $B(Z \rightarrow b\bar{b})$.

⁵ ABREU 97E value listed is for an assumed $\tau_{B_c} = 0.4$ ps and improves to 1.6×10^{-4} for $\tau_{B_c} = 1.4$ ps.

⁶ BARATE 97H reports $B(Z \rightarrow B_c X)/B(Z \rightarrow qq) \cdot B(B_c \rightarrow J/\psi(1S)\ell\nu_\ell) < 5.2 \times 10^{-5}$ at 90%CL. We rescale to our PDG 96 values of $B(Z \rightarrow b\bar{b})$. A $B_c^+ \rightarrow J/\psi(1S)\mu^+\nu_\mu$ candidate event is found, compared to all the known background sources 2×10^{-3} , which gives $m_{B_c} = 5.96^{+0.25}_{-0.19}$ GeV and $\tau_{B_c} = 1.77 \pm 0.17$ ps.

 $\Gamma(J/\psi(1S)\tau^+\nu_\tau)/\Gamma(J/\psi(1S)\mu^+\nu_\mu)$ **Γ_3/Γ_2**

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
0.52 ± 0.26 OUR AVERAGE		Error includes scale factor of 1.3.		
$0.17^{+0.18}_{-0.17} \pm 0.28$		¹ HAYRAPETY...25K	CMS	pp at 13 TeV
$0.71 \pm 0.17 \pm 0.18$		¹ AAIJ	18C LHCb	pp at 7, 8 TeV

¹ Uses $\tau^+ \rightarrow \mu^+\nu_\mu\bar{\nu}_\tau$ mode to obtain the ratio value.

 $\Gamma(J/\psi(1S)\pi^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)$ **$\Gamma_4/\Gamma \times B$**

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
seen		¹ AABOUD	21 ATLS	pp at 8 TeV
seen		² AAIJ	15M LHCb	pp at 8 TeV
seen		³ KHACHATRY...15AA	CMS	pp at 7 TeV
seen		AALTONEN	13 CDF	$p\bar{p}$ at 1.96 TeV

seen		⁴ AAIJ	12AV	LHCB	pp at 7 TeV
seen		AALTONEN	08M	CDF	$p\bar{p}$ at 1.96 TeV
seen		ABAZOV	08T	D0	$p\bar{p}$ at 1.96 TeV
• • • We do not use the following data for averages, fits, limits, etc. • • •					
$<2.4 \times 10^{-4}$	90	⁵ ACKERSTAFF	98O	OPAL	$e^+e^- \rightarrow Z$
$<3.4 \times 10^{-4}$	90	⁶ ABREU	97E	DLPH	$e^+e^- \rightarrow Z$
$<8.2 \times 10^{-5}$	90	⁷ BARATE	97H	ALEP	$e^+e^- \rightarrow Z$
$<2.0 \times 10^{-5}$	95	⁸ ABE	96R	CDF	$p\bar{p}$ 1.8 TeV
¹ AABOUD 21 reports a measurement of $B(B_c^+ \rightarrow J/\psi \pi^+) / B(B^+ \rightarrow J/\psi K^+)$ $\cdot f_c/f_u = (0.34 \pm 0.04_{-0.02}^{+0.06} \pm 0.01) \%$, at $p_T > 13$ GeV and $ y < 2.3$.					
² AAIJ 15M reports a measurement of $B(B_c^+ \rightarrow J/\psi \pi^+) / B(B^+ \rightarrow J/\psi K^+) \cdot f_c/f_u$ $= (0.683 \pm 0.018 \pm 0.009) \%$ at $p_T(B) < 20$ GeV and $2.0 < y(B) < 4.5$.					
³ KHACHATRYAN 15AA reports a measurement of $B(B_c^+ \rightarrow J/\psi \pi^+) / B(B^+ \rightarrow J/\psi K^+) \cdot f_c/f_u$ $= (0.48 \pm 0.05 \pm 0.03 \pm 0.05) \%$, at $p_T > 15$ GeV and $ \eta(B) < 1.6$.					
⁴ AAIJ 12AV reports a measurement of $B(B_c^+ \rightarrow J/\psi \pi^+) / B(B^+ \rightarrow J/\psi K^+) f_c/f_u$ $= (0.68 \pm 0.10 \pm 0.03 \pm 0.05) \%$ at $p_T(B) > 4$ GeV and $2.5 < \eta(B) < 4.5$.					
⁵ ACKERSTAFF 98O reports $B(Z \rightarrow B_c X) / B(Z \rightarrow qq) \times B(B_c \rightarrow J/\psi(1S)\pi^+) < 1.06 \times 10^{-4}$ at 90%CL. We rescale to our PDG 98 values of $B(Z \rightarrow b\bar{b})$.					
⁶ ABREU 97E value listed is for an assumed $\tau_{B_c} = 0.4$ ps and improves to 2.7×10^{-4} for $\tau_{B_c} = 1.4$ ps.					
⁷ BARATE 97H reports $B(Z \rightarrow B_c X) / B(Z \rightarrow qq) \cdot B(B_c \rightarrow J/\psi(1S)\pi) < 3.6 \times 10^{-5}$ at 90%CL. We rescale to our PDG 96 values of $B(Z \rightarrow b\bar{b})$.					
⁸ ABE 96R reports $B(b \rightarrow B_c X) / B(b \rightarrow B^+ X) \cdot B(B_c^+ \rightarrow J/\psi(1S)\pi^+) / B(B^+ \rightarrow J/\psi(1S)K^+) < 0.053$ at 95%CL for $\tau_{B_c} = 0.8$ ps. It changes from 0.15 to 0.04 for $0.17 \text{ ps} < \tau_{B_c} < 1.6$ ps. We rescale to our PDG 96 values of $B(b \rightarrow B^+) = 0.378 \pm 0.022$ and $B(B^+ \rightarrow J/\psi(1S)K^+) = 0.00101 \pm 0.00014$.					

$\Gamma(J/\psi(1S)\pi^+)/\Gamma(J/\psi(1S)\mu^+\nu_\mu)$ Γ_4/Γ_2

VALUE (units 10^{-2})	DOCUMENT ID	TECN	COMMENT
$4.69 \pm 0.28 \pm 0.46$	¹ AAIJ	14W	LHCB pp at 7 TeV
¹ AAIJ 14W reports also a measurement $B(B_c^+ \rightarrow J/\psi \pi^+) / B(B_c^+ \rightarrow J/\psi \mu^+ \nu_\mu) = 0.271 \pm 0.016 \pm 0.016$ in the region $m_{J/\psi \mu^+} > 5.3$ GeV.			

$\Gamma(J/\psi(1S)K^+)/\Gamma(J/\psi(1S)\pi^+)$ Γ_5/Γ_4

VALUE	EVTS	DOCUMENT ID	TECN	COMMENT
$0.079 \pm 0.007 \pm 0.003$		AAIJ	16AF	LHCB pp at 7, 8 TeV
• • • We do not use the following data for averages, fits, limits, etc. • • •				
$0.069 \pm 0.019 \pm 0.005$	50	AAIJ	13BY	LHCB Repl. by AAIJ 16AF

$\Gamma(J/\psi(1S)\pi^+\pi^0)/\Gamma(J/\psi(1S)\pi^+)$ Γ_6/Γ_4

VALUE	DOCUMENT ID	TECN	COMMENT
$2.80 \pm 0.15 \pm 0.19$	AAIJ	24L	LHCB pp at 7, 8, 13 TeV

$\Gamma(J/\psi(1S)\pi^+\pi^+\pi^-)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)$ $\Gamma_7/\Gamma \times B$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
seen		AAIJ	12Y	LHCB pp at 7 TeV

• • • We do not use the following data for averages, fits, limits, etc. • • •

$<5.7 \times 10^{-4}$ 90 ¹ ABREU 97E DLPH $e^+e^- \rightarrow Z$

¹ ABREU 97E value listed is independent of $0.4 \text{ ps} < \tau_{B_c} < 1.4 \text{ ps}$.

$\Gamma(J/\psi(1S)\pi^+\pi^+\pi^-)/\Gamma(J/\psi(1S)\pi^+)$ Γ_7/Γ_4

VALUE	DOCUMENT ID	TECN	COMMENT
2.4 ± 0.4 OUR AVERAGE			
$2.55 \pm 0.80 \pm 0.33^{+0.04}_{-0.01}$	KHACHATRY...15AA	CMS	pp at 7 TeV
$2.41 \pm 0.30 \pm 0.33$	AAIJ	12Y LHCb	pp at 7 TeV

$\Gamma(J/\psi(1S)K^+\pi^-\pi^+)/\Gamma(J/\psi(1S)K^+K^-\pi^+)$ Γ_8/Γ_{11}

VALUE	DOCUMENT ID	TECN	COMMENT
0.35 ± 0.06 ± 0.01	AAIJ	22P LHCb	pp at 7, 8, 13 TeV

$\Gamma(J/\psi(1S)K^+K^-K^+)/\Gamma(J/\psi(1S)K^+K^-\pi^+)$ Γ_9/Γ_{11}

VALUE	DOCUMENT ID	TECN	COMMENT
$(7.0 \pm 1.8 \pm 0.2) \times 10^{-2}$	AAIJ	22P LHCb	pp at 7, 8, 13 TeV

$\Gamma(J/\psi(1S)a_1(1260))/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)$ $\Gamma_{10}/\Gamma \times B$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<1.2 \times 10^{-3}$	90	¹ ACKERSTAFF 98O	OPAL	$e^+e^- \rightarrow Z$

¹ ACKERSTAFF 98O reports $B(Z \rightarrow B_c X)/B(Z \rightarrow qq) \times B(B_c \rightarrow J/\psi(1S)a_1(1260)) < 5.29 \times 10^{-4}$ at 90%CL. We rescale to our PDG 98 values of $B(Z \rightarrow b\bar{b})$.

$\Gamma(J/\psi(1S)K^+K^-\pi^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)$ $\Gamma_{11}/\Gamma \times B$

VALUE	DOCUMENT ID	TECN	COMMENT
seen	¹ AAIJ	13CA LHCb	pp at 7, 8 TeV

¹ A signal yield of 78 ± 14 decays is reported with a significance of 6.2 standard deviations using an integrated luminosity of 3 fb^{-1} data.

$\Gamma(J/\psi(1S)K^+K^-\pi^+)/\Gamma(J/\psi(1S)\pi^+)$ Γ_{11}/Γ_4

VALUE	DOCUMENT ID	TECN	COMMENT
0.53 ± 0.10 ± 0.05	¹ AAIJ	13CA LHCb	pp at 7, 8 TeV

¹ A signal yield of 78 ± 14 decays is reported with a significance of 6.2 standard deviations using an integrated luminosity of 3 fb^{-1} data.

$\Gamma(J/\psi(1S)K^+K^-\pi^+)/\Gamma(J/\psi(1S)\pi^+\pi^+\pi^-)$ Γ_{11}/Γ_7

VALUE	DOCUMENT ID	TECN	COMMENT
0.185 ± 0.013 ± 0.006	AAIJ	22P LHCb	pp at 7, 8, 13 TeV

$\Gamma(J/\psi(1S)\pi^+\pi^+\pi^+\pi^-\pi^-)/\Gamma(J/\psi(1S)\pi^+)$ Γ_{12}/Γ_4

VALUE	DOCUMENT ID	TECN	COMMENT
1.74 ± 0.44 ± 0.24	¹ AAIJ	14P LHCb	pp at 7, 8 TeV

¹ A signal yield of 32 ± 8 decays is reported with a significance of 4.5 standard deviations.

$\Gamma(\psi(2S)\pi^+)/\Gamma(J/\psi(1S)\pi^+)$ Γ_{13}/Γ_4

VALUE	DOCUMENT ID	TECN	COMMENT
0.254 ± 0.018 ± 0.006	^{1,2} AAIJ	24W LHCb	pp at 7, 8 and 13 TeV

• • • We do not use the following data for averages, fits, limits, etc. • • •

$0.268 \pm 0.032 \pm 0.007 \pm 0.006$ ^{3,4} AAIJ 15AY LHCB pp at 7, 8 TeV
 $0.250 \pm 0.068 \pm 0.014 \pm 0.006$ ³ AAIJ 13AMLHCB Repl. by AAIJ 15AY

¹ The last uncertainty includes the uncertainties on the branching fractions of the leptonic J/ψ and $\psi(2S)$ decays (± 0.005).

² Supersedes AAIJ 15AY.

³ The last uncertainty is due to the uncertainty of the $B(\psi(2S) \rightarrow \mu^+ \mu^-)/B(J/\psi \rightarrow \mu^+ \mu^-)$ ratio measurement.

⁴ Replaced by AAIJ 24W.

$\Gamma(\psi(2S)\pi^+)/\Gamma(J/\psi(1S)\pi^+\pi^-\pi^-)$ Γ_{13}/Γ_7

VALUE	DOCUMENT ID	TECN	COMMENT
$(3.5 \pm 0.6 \pm 0.2) \times 10^{-2}$	AAIJ	22P	LHCB pp at 7, 8, 13 TeV

$\Gamma(\psi(2S)\pi^+\pi^-\pi^+)/\Gamma(J/\psi(1S)\pi^+\pi^+\pi^-)$ Γ_{15}/Γ_7

VALUE	DOCUMENT ID	TECN	COMMENT
$(1.9 \pm 0.4 \pm 0.1) \times 10^{-2}$	AAIJ	22P	LHCB pp at 7, 8, 13 TeV

$\Gamma(\psi(2S)K^+K^-\pi^+)/\Gamma(J/\psi(1S)K^+K^-\pi^+)$ Γ_{16}/Γ_{11}

VALUE	DOCUMENT ID	TECN	COMMENT
$(3.7 \pm 1.2 \pm 0.1) \times 10^{-2}$	AAIJ	22P	LHCB pp at 7, 8, 13 TeV

$\Gamma(J/\psi(1S)D^0K^+)/\Gamma(J/\psi(1S)\pi^+)$ Γ_{17}/Γ_4

VALUE	DOCUMENT ID	TECN	COMMENT
$0.432 \pm 0.136 \pm 0.028$	AAIJ	17L	LHCB pp at 7, 8 TeV

$\Gamma(J/\psi(1S)D^*(2007)^0K^+)/\Gamma(J/\psi(1S)D^0K^+)$ Γ_{18}/Γ_{17}

VALUE	DOCUMENT ID	TECN	COMMENT
$5.1 \pm 1.8 \pm 0.4$	AAIJ	17L	LHCB pp at 7, 8 TeV

$\Gamma(J/\psi(1S)D^*(2010)^+K^{*0})/\Gamma(J/\psi(1S)D^0K^+)$ Γ_{19}/Γ_{17}

VALUE	DOCUMENT ID	TECN	COMMENT
$2.10 \pm 1.08 \pm 0.34$	AAIJ	17L	LHCB pp at 7, 8 TeV

$\Gamma(J/\psi(1S)D^+K^{*0})/\Gamma(J/\psi(1S)D^0K^+)$ Γ_{20}/Γ_{17}

VALUE	DOCUMENT ID	TECN	COMMENT
$0.63 \pm 0.39 \pm 0.08$	AAIJ	17L	LHCB pp at 7, 8 TeV

$\Gamma(J/\psi(1S)D_s^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)$ $\Gamma_{21}/\Gamma \times B$

VALUE (units 10^{-6})	DOCUMENT ID	TECN	COMMENT
$6.7 \pm 0.8 \pm 0.1$	¹ AAIJ	24F	LHCB pp at 7, 8, 13 TeV

¹ AAIJ 24F reports $[\Gamma(B_c^+ \rightarrow J/\psi(1S)D_s^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] = (1.63 \pm 0.15 \pm 0.13) \times 10^{-5}$ which we multiply by our best (shown rounded) value $B(\bar{b} \rightarrow B^+ \text{ at } Z) = (40.8 \pm 0.7) \times 10^{-2}$. Our first error is their experiment's error and our second error is the systematic error from using our best (shown rounded) value.

$$\Gamma(J/\psi(1S)D_s^+)/\Gamma(J/\psi(1S)\pi^+) \quad \Gamma_{21}/\Gamma_4$$

VALUE	DOCUMENT ID	TECN	COMMENT
2.8 ± 0.4 OUR AVERAGE			
2.76 ± 0.33 ± 0.33	¹ AAD	220 ATLS	pp at 13 TeV
2.90 ± 0.57 ± 0.24	AAIJ	13AS LHCB	pp at 7, 8 TeV
• • • We do not use the following data for averages, fits, limits, etc. • • •			
3.8 ± 1.1 ± 0.4	AAD	16H ATLS	pp at 7, 8 TeV

¹Supersedes the measurement of AAD 16H.

$$\Gamma(J/\psi(1S)D_s^{*+})/\Gamma(J/\psi(1S)\pi^+) \quad \Gamma_{22}/\Gamma_4$$

VALUE	DOCUMENT ID	TECN	COMMENT
5.33 ± 0.61 ± 0.74	AAD	220 ATLS	pp at 13 TeV
• • • We do not use the following data for averages, fits, limits, etc. • • •			
10.4 ± 3.1 ± 1.6	AAD	16H ATLS	Repl. by AAD 220

$$\Gamma(J/\psi(1S)D_s^{*+})/\Gamma(J/\psi(1S)D_s^+) \quad \Gamma_{22}/\Gamma_{21}$$

VALUE	DOCUMENT ID	TECN	COMMENT
1.95 ± 0.16 OUR AVERAGE			
1.91 ± 0.20 ± 0.07	AAIJ	24F LHCB	pp at 7, 8, 13 TeV
1.93 ± 0.24 ± 0.09	¹ AAD	220 ATLS	pp at 13 TeV
2.37 ± 0.56 ± 0.10	AAIJ	13AS LHCB	pp at 7, 8 TeV
• • • We do not use the following data for averages, fits, limits, etc. • • •			
2.8 $^{+1.2}_{-0.8}$ ± 0.3	AAD	16H ATLS	pp at 7, 8 TeV

¹Supersedes the measurement of AAD 16H.

$$\Gamma(J/\psi(1S)p\bar{p}\pi^+)/\Gamma(J/\psi(1S)\pi^+) \quad \Gamma_{23}/\Gamma_4$$

VALUE	DOCUMENT ID	TECN	COMMENT
0.143 $^{+0.041}_{-0.036}$	AAIJ	14AQ LHCB	pp at 7, 8 TeV

$$\Gamma(\chi_{c0}\pi^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{24}/\Gamma \times B$$

VALUE (units 10^{-6})	DOCUMENT ID	TECN	COMMENT
24.0 $^{+8.6}_{-7.6}$ ± 0.4	^{1,2} AAIJ	16AT LHCB	pp at 7 and 8 TeV

¹AAIJ 16AT reports $[\Gamma(B_c^+ \rightarrow \chi_{c0}\pi^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] \times [\Gamma(\bar{b} \rightarrow B^+ \text{ at } Z)/\Gamma_{\text{total}}] = (9.8^{+3.4}_{-3.0} \pm 0.8) \times 10^{-6}$ which we divide by our best (shown rounded) value $\Gamma(\bar{b} \rightarrow B^+ \text{ at } Z)/\Gamma_{\text{total}} = 0.408 \pm 0.007$. Our first error is their experiment's error and our second error is the systematic error from using our best (shown rounded) value.

²The significance of the observed signal is 4.0 standard deviations.

$$\Gamma(\chi_{c1}\pi^+)/\Gamma(\chi_{c2}\pi^+) \quad \Gamma_{25}/\Gamma_{26}$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
< 0.49	90	AAIJ	24I LHCB	pp at 7, 8 and 13 TeV

$$\Gamma(\chi_{c2}\pi^+)/\Gamma(J/\psi(1S)\pi^+) \quad \Gamma_{26}/\Gamma_4$$

VALUE	DOCUMENT ID	TECN	COMMENT
0.37 ± 0.06 ± 0.022	¹ AAIJ	24I LHCB	pp at 7, 8 and 13 TeV

¹The last uncertainty includes the knowledge of the $\chi_{c2} \rightarrow J/\psi\gamma$ branching fraction (± 0.01).

$$\Gamma(\chi_{c1}(3872)\pi^+, \chi_{c1}(3872) \rightarrow J/\psi\pi^+\pi^-)/\Gamma(\psi(2S)\pi^+, \psi(2S) \rightarrow J/\psi\pi^+\pi^-) \quad \Gamma_{27}/\Gamma_{14}$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<5 \times 10^{-2}$	90	AAIJ	25V	LHCB pp at 7, 8, 13 TeV

$$\Gamma(p\bar{p}\pi^+)/\Gamma_{\text{total}} \quad \Gamma_{28}/\Gamma$$

VALUE	DOCUMENT ID	TECN	COMMENT
not seen	¹ AAIJ	16K	LHCB pp at 7, 8 TeV

¹ Measures the ratio $(f_c/f_u) \times B(B_c^+ \rightarrow p\bar{p}\pi^+) < 3.6 \times 10^{-8}$ at 95% CL, in the region $m(p\bar{p}) < 2.85 \text{ GeV}/c^2$, where f_c (f_u) represents the fragmentation fraction of the b -quark into the B_c^+ (B_u^+) meson.

$$\Gamma(D^0 K^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{29}/\Gamma \times B$$

VALUE (units 10^{-7})	DOCUMENT ID	TECN	COMMENT
$3.8^{+1.2}_{-1.0} \pm 0.1$	¹ AAIJ	17AG	LHCB pp at 7, 8 TeV

¹ AAIJ 17AG reports $[\Gamma(B_c^+ \rightarrow D^0 K^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] = (9.3^{+2.8}_{-2.5} \pm 0.6) \times 10^{-7}$ which we multiply by our best (shown rounded) value $B(\bar{b} \rightarrow B^+ \text{ at } Z) = (40.8 \pm 0.7) \times 10^{-2}$. Our first error is their experiment's error and our second error is the systematic error from using our best (shown rounded) value.

$$\Gamma(D^0 \pi^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{30}/\Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<1.6 \times 10^{-7}$	95	¹ AAIJ	17AG	LHCB pp at 7, 8 TeV

¹ AAIJ 17AG reports $[\Gamma(B_c^+ \rightarrow D^0 \pi^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] < 3.9 \times 10^{-7}$ which we multiply by our best (shown rounded) value $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$.

$$\Gamma(D^{*0} \pi^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{31}/\Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<4 \times 10^{-7}$	95	¹ AAIJ	17AG	LHCB pp at 7, 8 TeV

¹ AAIJ 17AG reports $[\Gamma(B_c^+ \rightarrow D^{*0} \pi^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] < 1.1 \times 10^{-6}$ which we multiply by our best (shown rounded) value $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$.

$$\Gamma(D^{*0} K^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{32}/\Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<4 \times 10^{-7}$	95	¹ AAIJ	17AG	LHCB pp at 7, 8 TeV

¹ AAIJ 17AG reports $[\Gamma(B_c^+ \rightarrow D^{*0} K^+)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] < 1.1 \times 10^{-6}$ which we multiply by our best (shown rounded) value $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$.

$$\Gamma(D_s^+ \bar{D}^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{33}/\Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<1.4 \times 10^{-7}$	90	¹ AAIJ	18P	LHCB pp at 7, 8 TeV

¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D_s^+ \bar{D}^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)]$
 $/ [B(B^+ \rightarrow \bar{D}^0 D^+)] < 0.9 \times 10^{-3}$ which we multiply by our best (shown rounded)
 values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$\Gamma(D_s^+ \bar{D}^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)$ $\Gamma_{34}/\Gamma \times B$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$< 6 \times 10^{-8}$	90	¹ AAIJ	18P LHCB	pp at 7, 8 TeV

¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D_s^+ \bar{D}^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)]$
 $/ [B(B^+ \rightarrow \bar{D}^0 D^+)] < 3.7 \times 10^{-4}$ which we multiply by our best (shown rounded)
 values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$\Gamma(D^+ \bar{D}^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)$ $\Gamma_{35}/\Gamma \times B$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$< 3.0 \times 10^{-6}$	90	¹ AAIJ	18P LHCB	pp at 7, 8 TeV

¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D^+ \bar{D}^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)]$
 $/ [B(B^+ \rightarrow \bar{D}^0 D^+)] < 1.9 \times 10^{-2}$ which we multiply by our best (shown rounded)
 values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$\Gamma(D^+ \bar{D}^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)$ $\Gamma_{36}/\Gamma \times B$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
• • • We do not use the following data for averages, fits, limits, etc. • • •				

$< 1.9 \times 10^{-6}$ 90 ¹ AAIJ 18P LHCB Repl. by AAIJ 21AF

¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D^+ \bar{D}^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)]$
 $/ [B(B^+ \rightarrow \bar{D}^0 D^+)] < 1.2 \times 10^{-2}$ which we multiply by our best (shown rounded)
 values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$[\Gamma(D_s^{*+} \bar{D}^0) + \Gamma(D_s^+ \bar{D}^{*0}(2007^0))]/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)$ $(\Gamma_{37} + \Gamma_{38})/\Gamma \times B$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
• • • We do not use the following data for averages, fits, limits, etc. • • •				

$< 4 \times 10^{-7}$ 90 ¹ AAIJ 18P LHCB Repl. by AAIJ 21AF

¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D_s^{*+} \bar{D}^0) + \Gamma(B_c^+ \rightarrow D_s^+ \bar{D}^{*0}(2007^0))]/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] / [B(B^+ \rightarrow \bar{D}^0 D^+)] < 2.8 \times 10^{-3}$ which we multiply
 by our best (shown rounded) values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$[\Gamma(D_s^{*+} D^0) + \Gamma(D_s^+ D^{*0}(2007^0))]/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)$ $(\Gamma_{39} + \Gamma_{40})/\Gamma \times B$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
• • • We do not use the following data for averages, fits, limits, etc. • • •				

$< 5 \times 10^{-7}$ 90 ¹ AAIJ 18P LHCB Repl. by AAIJ 21AF

¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D_s^{*+} D^0) + \Gamma(B_c^+ \rightarrow D_s^+ D^{*0}(2007^0))]/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] / [B(B^+ \rightarrow \bar{D}^0 D^+)] < 3.0 \times 10^{-3}$ which we multiply
 by our best (shown rounded) values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$$\Gamma(D^*(2010)^+\bar{D}^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{41}/\Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<6.2 \times 10^{-3}$	90	¹ BARATE	98Q ALEP	$e^+e^- \rightarrow Z$

¹ BARATE 98Q reports $B(Z \rightarrow B_c X) \times B(B_c \rightarrow D^*(2010)^+\bar{D}^0) < 1.9 \times 10^{-3}$ at 90%CL. We rescale to our PDG 98 values of $B(Z \rightarrow b\bar{b})$.

$$[\Gamma(D^*(2010)^+\bar{D}^0, D^{*+} \rightarrow D^+\pi^0/\gamma) + \Gamma(D^+\bar{D}^*(2007)^0)]/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad (\Gamma_{42} + \Gamma_{43})/\Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
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• • • We do not use the following data for averages, fits, limits, etc. • • •

$<9 \times 10^{-6}$ 90 ¹ AAIJ 18P LHCb Repl. by AAIJ 21AF

¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D^*(2010)^+\bar{D}^0, D^{*+} \rightarrow D^+\pi^0/\gamma) + \Gamma(B_c^+ \rightarrow D^+\bar{D}^*(2007)^0)]/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] / [B(B^+ \rightarrow \bar{D}^0 D^+)] < 5.5 \times 10^{-2}$ which we multiply by our best (shown rounded) values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$$[\Gamma(D^*(2010)^+D^0, D^{*+} \rightarrow D^+\pi^0/\gamma) + \Gamma(D^+D^*(2007)^0)]/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad (\Gamma_{45} + \Gamma_{46})/\Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
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• • • We do not use the following data for averages, fits, limits, etc. • • •

$<3.4 \times 10^{-6}$ 90 ¹ AAIJ 18P LHCb Repl. by AAIJ 21AF

¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D^*(2010)^+D^0, D^{*+} \rightarrow D^+\pi^0/\gamma) + \Gamma(B_c^+ \rightarrow D^+D^*(2007)^0)]/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] / [B(B^+ \rightarrow \bar{D}^0 D^+)] < 2.2 \times 10^{-2}$ which we multiply by our best (shown rounded) values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$$\Gamma(D_s^{*+}\bar{D}^*(2007)^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{47}/\Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
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• • • We do not use the following data for averages, fits, limits, etc. • • •

$<1.7 \times 10^{-6}$ 90 ¹ AAIJ 18P LHCb Repl. by AAIJ 21AF

¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D_s^{*+}\bar{D}^*(2007)^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] / [B(B^+ \rightarrow \bar{D}^0 D^+)] < 1.1 \times 10^{-2}$ which we multiply by our best (shown rounded) values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$$\Gamma(D_s^{*+}D^*(2007)^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{48}/\Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
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• • • We do not use the following data for averages, fits, limits, etc. • • •

$<3.1 \times 10^{-6}$ 90 ¹ AAIJ 18P LHCb Repl. by AAIJ 21AF

¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D_s^{*+}D^*(2007)^0)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] / [B(B^+ \rightarrow \bar{D}^0 D^+)] < 2.0 \times 10^{-2}$ which we multiply by our best (shown rounded) values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$$\Gamma(D^*(2010)^+ \bar{D}^*(2007)^0) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{49} / \Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
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• • • We do not use the following data for averages, fits, limits, etc. • • •

$< 1.0 \times 10^{-4}$	90	¹ AAIJ	18P LHCb	Repl. by AAIJ 21AF
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¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D^*(2010)^+ \bar{D}^*(2007)^0) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] / [B(B^+ \rightarrow \bar{D}^0 D^+)] < 6.5 \times 10^{-1}$ which we multiply by our best (shown rounded) values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$$\Gamma(D^*(2010)^+ D^*(2007)^0) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{50} / \Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
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• • • We do not use the following data for averages, fits, limits, etc. • • •

$< 2.0 \times 10^{-5}$	90	¹ AAIJ	18P LHCb	Repl. by AAIJ 21AF
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¹ AAIJ 18P reports $[\Gamma(B_c^+ \rightarrow D^*(2010)^+ D^*(2007)^0) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] / [B(B^+ \rightarrow \bar{D}^0 D^+)] < 1.3 \times 10^{-1}$ which we multiply by our best (shown rounded) values $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$, $B(B^+ \rightarrow \bar{D}^0 D^+) = 3.8 \times 10^{-4}$.

$$\Gamma(D^+ K^{*0}) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{51} / \Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
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$< 2.0 \times 10^{-7}$	90	¹ AAIJ	13R LHCb	pp at 7 TeV
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¹ AAIJ 13R reports $[\Gamma(B_c^+ \rightarrow D^+ K^{*0}) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] < 0.5 \times 10^{-6}$ which we multiply by our best (shown rounded) value $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$.

$$\Gamma(D^+ \bar{K}^{*0}) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{52} / \Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
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$< 1.6 \times 10^{-7}$	90	¹ AAIJ	13R LHCb	pp at 7 TeV
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¹ AAIJ 13R reports $[\Gamma(B_c^+ \rightarrow D^+ \bar{K}^{*0}) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] < 0.4 \times 10^{-6}$ which we multiply by our best (shown rounded) value $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$.

$$\Gamma(D_s^+ K^{*0}) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{53} / \Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
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$< 2.9 \times 10^{-7}$	90	¹ AAIJ	13R LHCb	pp at 7 TeV
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¹ AAIJ 13R reports $[\Gamma(B_c^+ \rightarrow D_s^+ K^{*0}) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] < 0.7 \times 10^{-6}$ which we multiply by our best (shown rounded) value $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$.

$$\Gamma(D_s^+ \bar{K}^{*0}) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{54} / \Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
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$< 4 \times 10^{-7}$	90	¹ AAIJ	13R LHCb	pp at 7 TeV
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¹ AAIJ 13R reports $[\Gamma(B_c^+ \rightarrow D_s^+ \bar{K}^{*0}) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] < 1.1 \times 10^{-6}$ which we multiply by our best (shown rounded) value $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$.

$$\Gamma(D_s^+ \phi) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{55} / \Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$< 3.3 \times 10^{-7}$	90	¹ AAIJ	13R LHCB	pp at 7 TeV

¹ AAIJ 13R reports $[\Gamma(B_c^+ \rightarrow D_s^+ \phi) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] < 0.8 \times 10^{-6}$ which we multiply by our best (shown rounded) value $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$.

$$\Gamma(K^+ K^0) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{56} / \Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$< 4.6 \times 10^{-7}$	90	¹ AAIJ	13BS LHCB	pp at 7 TeV

¹ Derived from $\Gamma(K^+ K^0) / \Gamma \times B(\bar{b} \rightarrow B_c) / (B(B^+ \rightarrow K^0 \pi^+) B(\bar{b} \rightarrow B^+)) < 5.8\%$ at 90% CL using normalization mode $B(B^+ \rightarrow K^0 \pi^+) = (23.97 \pm 0.53 \pm 0.71) \times 10^{-6}$ and assuming a B production ratio $f(\bar{b} \rightarrow B_u^+) = 0.33$.

$$\Gamma(B_s^0 \pi^+ / B(\bar{b} \rightarrow B_s)) / \Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{59} / \Gamma \times B$$

VALUE (units 10^{-3})	DOCUMENT ID	TECN	COMMENT
$2.37 \pm 0.31 \pm 0.11^{+0.17}_{-0.13}$	¹ AAIJ	13BU LHCB	pp at 7, 8 TeV

¹ The last uncertainty is due to the uncertainty of the B_c^+ lifetime measurement.

$$\Gamma(D_s^+ \bar{D}^0) / \Gamma_{\text{total}} \quad \Gamma_{33} / \Gamma$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$< 7.2 \times 10^{-4}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c) / B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19AI.

$$\Gamma(D_s^+ D^0) / \Gamma_{\text{total}} \quad \Gamma_{34} / \Gamma$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$< 3.0 \times 10^{-4}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c) / B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19AI.

$$\Gamma(D^+ \bar{D}^0) / \Gamma_{\text{total}} \quad \Gamma_{35} / \Gamma$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$< 1.9 \times 10^{-4}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c) / B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19AI.

$$\Gamma(D^+ D^0) / \Gamma_{\text{total}} \quad \Gamma_{36} / \Gamma$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$< 1.4 \times 10^{-4}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c) / B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19AI.

$$\Gamma(D_s^{*+} \bar{D}^0) / \Gamma_{\text{total}} \quad \Gamma_{37} / \Gamma$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$< 5.3 \times 10^{-4}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c) / B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19AI.

$\Gamma(D_s^+ \bar{D}^*(2007)^0)/\Gamma_{\text{total}}$ Γ_{38}/Γ

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<4.6 \times 10^{-4}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c)/B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19Al. $\Gamma(D_s^{*+} D^0)/\Gamma_{\text{total}}$ Γ_{39}/Γ

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<0.9 \times 10^{-3}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c)/B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19Al. $\Gamma(D_s^+ D^*(2007)^0)/\Gamma_{\text{total}}$ Γ_{40}/Γ

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<6.6 \times 10^{-4}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c)/B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19Al. $\Gamma(D^*(2010)^+ \bar{D}^0)/\Gamma_{\text{total}}$ Γ_{41}/Γ

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<3.8 \times 10^{-4}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c)/B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19Al. $\Gamma(D^*(2007)^+ D^0)/\Gamma_{\text{total}}$ Γ_{44}/Γ

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<2.0 \times 10^{-4}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c)/B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19Al. $\Gamma(D^+ \bar{D}^*(2007)^0)/\Gamma_{\text{total}}$ Γ_{43}/Γ

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<6.5 \times 10^{-4}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c)/B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19Al. $\Gamma(D^+ D^*(2007)^0)/\Gamma_{\text{total}}$ Γ_{46}/Γ

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<3.7 \times 10^{-4}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c)/B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19Al. $\Gamma(D_s^{*+} \bar{D}^*(2007)^0)/\Gamma_{\text{total}}$ Γ_{47}/Γ

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<1.3 \times 10^{-3}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c)/B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19Al. $\Gamma(D_s^{*+} D^*(2007)^0)/\Gamma_{\text{total}}$ Γ_{48}/Γ

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<1.3 \times 10^{-3}$	90	¹ AAIJ	21AF LHCB	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c)/B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19Al.

$$\Gamma(D^*(2010)^+ \bar{D}^*(2007)^0)/\Gamma_{\text{total}} \quad \Gamma_{49}/\Gamma$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<1.0 \times 10^{-3}$	90	¹ AAIJ	21AF LHCb	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c)/B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19AI.

$$\Gamma(D^*(2010)^+ D^*(2007)^0)/\Gamma_{\text{total}} \quad \Gamma_{50}/\Gamma$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<7.7 \times 10^{-4}$	90	¹ AAIJ	21AF LHCb	pp at 13 TeV

¹ Uses $B(\bar{b} \rightarrow B_c)/B(\bar{b} \rightarrow B^+) = 0.76\%$ determined by AAIJ 19AI.

$$\Gamma(\pi^+ B^{*0}, B^{*0} \rightarrow \mu^+ \mu^-)/\Gamma(J/\psi(1S)\pi^+) \quad \Gamma_{57}/\Gamma_4$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<3.8 \times 10^{-5}$	90	AAIJ	25Z LHCb	pp at 7, 8, 13 TeV

$$\Gamma(\pi^+ B_s^{*0}, B_s^{*0} \rightarrow \mu^+ \mu^-)/\Gamma(J/\psi(1S)\pi^+) \quad \Gamma_{58}/\Gamma_4$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<5.0 \times 10^{-5}$	90	AAIJ	25Z LHCb	pp at 7, 8, 13 TeV

$$\Gamma(B_s^0 \pi^+)/\Gamma(J/\psi(1S)\pi^+) \quad \Gamma_{60}/\Gamma_4$$

VALUE	DOCUMENT ID	TECN	COMMENT
$91 \pm 10 \pm 8.5$	¹ AAIJ	23M LHCb	pp at 13 TeV

¹ The B_s^0 mesons are reconstructed via the decays $B_s^0 \rightarrow J/\psi \phi$ and $B_s^0 \rightarrow D_s^- \pi^+$. The third uncertainty includes systematic (± 8) and imprecise knowledge of the branching fractions (± 3).

$$\Gamma(\pi^+ \mu^+ \mu^-)/\Gamma(J/\psi(1S)\pi^+) \quad \Gamma_{61}/\Gamma_4$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<2.1 \times 10^{-4}$	90	AAIJ	24W LHCb	pp at 7, 8 and 13 TeV

$$\Gamma(D_s^+ \mu^+ \mu^-)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z) \quad \Gamma_{62}/\Gamma \times B$$

VALUE	CL%	DOCUMENT ID	TECN	COMMENT
$<3.1 \times 10^{-8}$	90	¹ AAIJ	24F LHCb	pp at 7, 8, 13 TeV

¹ AAIJ 24F reports $[\Gamma(B_c^+ \rightarrow D_s^+ \mu^+ \mu^-)/\Gamma_{\text{total}} \times B(\bar{b} \rightarrow B_c \text{ at } Z)] / [B(\bar{b} \rightarrow B^+ \text{ at } Z)] < 7.5 \times 10^{-8}$ which we multiply by our best (shown rounded) value $B(\bar{b} \rightarrow B^+ \text{ at } Z) = 40.8 \times 10^{-2}$.

POLARIZATION IN B_c^+ DECAY

In decays involving two vector mesons, one can distinguish among the states in which meson polarizations are both longitudinal (L) or both are transverse and parallel ($\parallel$) or perpendicular ($\perp$) to each other with the parameters Γ_L/Γ , $\Gamma_\perp/\Gamma$, and the relative phases $\phi_\parallel$ and $\phi_\perp$. See the definitions in the note on “Polarization in B Decays” review in the B^0 Particle Listings.

$$\Gamma_L/\Gamma \text{ in } B_c^+ \Rightarrow J/\psi D_s^{*+}$$

VALUE	DOCUMENT ID	TECN	COMMENT
0.34 ± 0.09 OUR AVERAGE			
$0.30 \pm 0.10 \pm 0.04$	^{1,2} AAD	220 ATLS	pp at 13 TeV

0.48 ± 0.20 ³ AAIJ 13AS LHCb pp at 7, 8 TeV

• • • We do not use the following data for averages, fits, limits, etc. • • •

0.62 ± 0.24 ⁴ AAD 16H ATLAS pp at 7, 8 TeV

¹ Supersedes the measurement of AAD 16H.

² AAD 220 measures $1 - \Gamma_L/\Gamma = 0.70 \pm 0.10 \pm 0.04$.

³ AAIJ 13AS measures $1 - \Gamma_L/\Gamma = 0.52 \pm 0.20$.

⁴ AAD 16H measures $1 - \Gamma_L/\Gamma = 0.38 \pm 0.24$.

$A_P(B_c^+)$

$$A_P(B_c^+) = [\sigma(B_c^-) - \sigma(B_c^+)] / [\sigma(B_c^-) + \sigma(B_c^+)]$$

VALUE (units 10^{-2})	DOCUMENT ID	TECN	COMMENT
$=1.0 \pm 1.0$ OUR AVERAGE			
$-2.5 \pm 2.1 \pm 0.5$	¹ AAIJ	19AI LHCb	pp at 7 TeV
$-0.5 \pm 1.1 \pm 0.4$	¹ AAIJ	19AI LHCb	pp at 13 TeV
¹ Measured using B_c^+ semileptonic decays.			

B_c^+ REFERENCES

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